IVF/NSGA-III - In Vitro Fertilization method coupled to NSGA-III

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***Abstract*—The In Vitro Fertilization Genetic Algorithm - IVF/GA is a promising algorithm applicable to single-objective problems, especially complex and multimodal ones. Due to the balance between its exploration and exploitation phases, and its capability to avoid local optimal, we speculate that this method can also improve Many-Objective Evolutionary Algorithms. Thus, this work proposes the adaptation of the In Vitro Fertilization method to the Many-Objective approach, with a new collection, manipulation and transfer criteria, as well as its coupling to the Many-Objective Evolutionary Algorithm NSGA-III, called IVF/NSGA-III. We evaluated the efficacy of the proposals by comparing canonic NSGA-III with memetic IVF/NSGA-III, applied to Many-Objective benchmark DTLZ. The results show that IVF/NSGA-III outperformed canonical NSGA-III in most problems addressed. The results also indicate that this proposal is promising to improve MOEAs.**

***Index Terms*—In Vitro Fertilization method, Multi-Objective and Many-Objective Memetic Algorithms, Multi-Objective and Many-Objective Evolutionary Algorithms, Hybridism, Genetic Algorithms, Differential Evolution, NSGA-III**

I. Introduction

Several real-world problems have been modelled as optimization problems and solved with computational techniques [1]. In most cases, the problem requires the optimization of two or more objectives simultaneously; thus, the solution must consider the trade-off between different objectives [2]. These problems are known as Multi-objective Problems (MOP), and many algorithms have been developed to solve them.

Evolutionary multi-objective optimization (EMO) algorithms, inspired by the Darwinian principles, evolve a population of individuals representing the solutions for the problem to explore the space of possible solutions efficiently. NSGA-II and SPEA2 are examples of EMO widely used to solve MOPs. Meanwhile, the EMOs NSGA-III and MOEA/D deal with problems with many objectives, in other words, when the problem has more than three objectives [3].

Although same inspiration, evolutionary algorithms differ in how genetic information is manipulated and how individuals' adaptation criteria are implemented. Thus, there are families of algorithms such as Genetic Algorithms (GA), Evolutionary Strategies (ES), Genetic Programming (GP) and Differential Evolution (DE).

Despite the strategy employed in the EMO, all the algorithms try to control the evolution process balancing the convergence and diversity of the population to achieve good results. However, this may be a challenge in some real-world problems. Thus, many proposals appeal to the so-called hybridism, combining promising aspects of different algorithms to obtain a superior one.

Proposed in [4], In Vitro Fertilization module (IVF) has demonstrated promising results when coupled with several evolutionary algorithms for solving both single and multi-objective optimization problems, respectively, in [5] and [6].

Previous works show that IVF can improve the balance of exploration and exploitation during the evolutionary process, contributing to a better search space exploration and avoiding local optima.

Thus, this work investigates if the characteristics of IVF can also contribute to solve many-objective problems. Hence, we propose the In Vitro Fertilization NSGA-III (IVF/NSGA-III) by adapting and coupling the IVF to NSGA-III [7].

To this end, we introduced new criteria for collecting, manipulating, and transferring individuals. In the manipulation stage, we adopt perturbation based on Differential Evolution aims at approximating individuals to niches of interest.

Therefore, the main contributions of this paper are:

- A new hybrid evolutionary algorithm based on NSGA-III that, in general, outperforms the canonical one.
- An expansion of the In Vitro Fertilization stages adopting a modified Differential Evolution with beacons and adaptive scale factor for elitist local search.
- A strategy to reduce the number of parameters when applying In Vitro Fertilization module.
- An experimental study using a well-established benchmark in the context of many-objective problems.

The remainder of this paper is organized as follows. Section II presents the concepts many-objective problems, NSGA-III, Hybridism and In Vitro Fertilization Genetic Algorithm. Section III describes the IVF/NSGA-III proposal. Section IV exposes the experimentation design. Section V shows and discusses the results, and finally Section VI presents the conclusions and future work.

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II. Background and Related Work

A. Many-Objectives problem

When the optimization problem has more than one objective to be optimized, the problem is a MOP. If the problem has four or more objectives, such a problem is a many-objective problem. One reason for this distinction is the possibility of visualizing the search space surface until three dimensions. However, visualization is hard for more dimensions. Many-objective problems are particularly challenging for EMOs for some reasons: i) it is more challenging to apply dominance criteria; ii) computational cost to measure metrics such as diversity and hypervolume, and iii) recombination operation may be inefficient because high-dimensionality tends to pose solutions so far from each other, then recombination may generate a solution much different from the parents [7].

B. Non Dominated Sorting Genetic Algorithm - NSGA-III

In the context of many-objective problems, one of the most used algorithms is NSGA-III [7]. The NSGA-III was proposed following the framework introduced in NSGA-II but using a different strategy for the selection operator. Instead of applying crowding distance to ensure diversity in the population, as in the previous algorithm, NSGA-III implements the concept of reference points. After executing the standard selection by dominance, the algorithm associates the individuals to the reference points, creating some niches and using these niches to complement the next generation.

C. Differential Evolution

Differential evolution (DE) [8] is a family of Evolutionary Algorithms (EA) that have been successfully applied to problems with continuous search space. According to [9], DE variants perform common steps for an EA. But, unlike other EAs, DE operates over a difference vector to promote perturbation over the generations. Instead of directly changing a gene based on independent probability to produce a mutation, DE models individuals as vectors. Then, it selects a donor and a parent vector and recombines them to produce offspring. There are various mutation strategies, all of them are based on difference vectors weighted by a scaling factor F.

D. Hybridism in Evolutionary Algorithms

In general, hybridism combines promising aspects of different techniques to produce a better approach. Considering the context of EA, according to [10], hybridization can improve the performance, for instance, speeding up the convergence, increasing the quality of the solutions and enabling the incorporation of the EA algorithm as part of another system. In [4], the authors show how an auxiliary algorithm was able to improve a GA. In [8] there is a list of cases in which hybridism improved DE variations.

E. In Vitro Fertilization Genetic Algorithm - IVF/GA

Proposed in [4], IVF/GA is a Genetic Algorithm with a parallel and auxiliary module called In Vitro Fertilization module (IVF), as illustrated in Fig 1. IVF is responsible for mining genetic information from a part of the GA population, trying to recombine promising genetic material inspired by what happens with the "test tube babe".

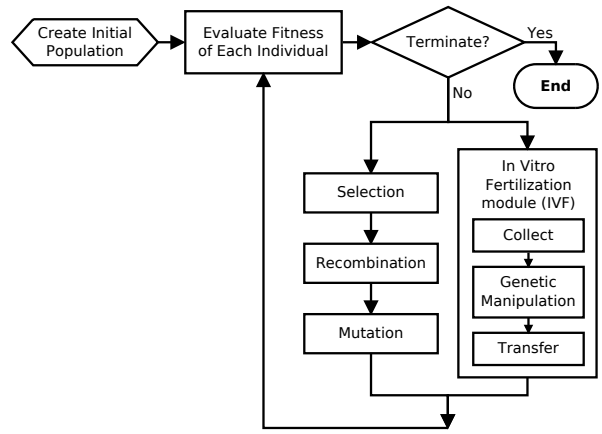

Fig. 1. In Vitro Fertilization method flow.

In practice, the IVF implements the following steps: Collect, Genetic Manipulation and Transfer. Collect stage defines the number and which individuals would be collected from the host population; Genetic Manipulation has two parts: 1) application of exploratory operators on the subpopulation collected, modifying partial or totally its genetic material; 2) execution of a recombination operation, by assigning the best individual as "father" and all the rest as "mothers". Finally, the Transfer stage defines how many and which individuals produced in the second stage must send back to the population of the GA or another Evolutionary Algorithm.

F. IVF module coupled to Evolutionary Algorithm

Previous works [5], [6] demonstrated the potential of coupling the IVF to well-established Multi-Objective Evolutionary Algorithms. The first proposed a hybrid algorithm combining IVF with NSGA-II. The results pointed to the approach IVF/NSGA-II being better than the NSGA-II, mainly in the complex and multimodal ZDT4 function. Later, IVF was incorporated into the GDE3 (an extension of DE) [11]. The new hybrid algorithm IVF/GDE3 outperformed the canonical GDE3 in convergence and efficacy in most tested benchmarks. Because of these promising findings, in this work we explore hybridism to propose a new hybrid Many-Objective Evolutionary Algorithm based on IVF, NSGA-III and Differential Evolution.

III. Approach

This session presents a proposal for adapting the In Vitro Fertilization method to the Many-Objective approach, precisely using the problem decomposition method, with niches associated with reference lines adopted in the NSGA-III.

Fig. 2 presents a summarized diagram of this hybridism proposal, with the In Vitro Fertilization method adapted to the many-objective approach and coupled to the NSGA-III.

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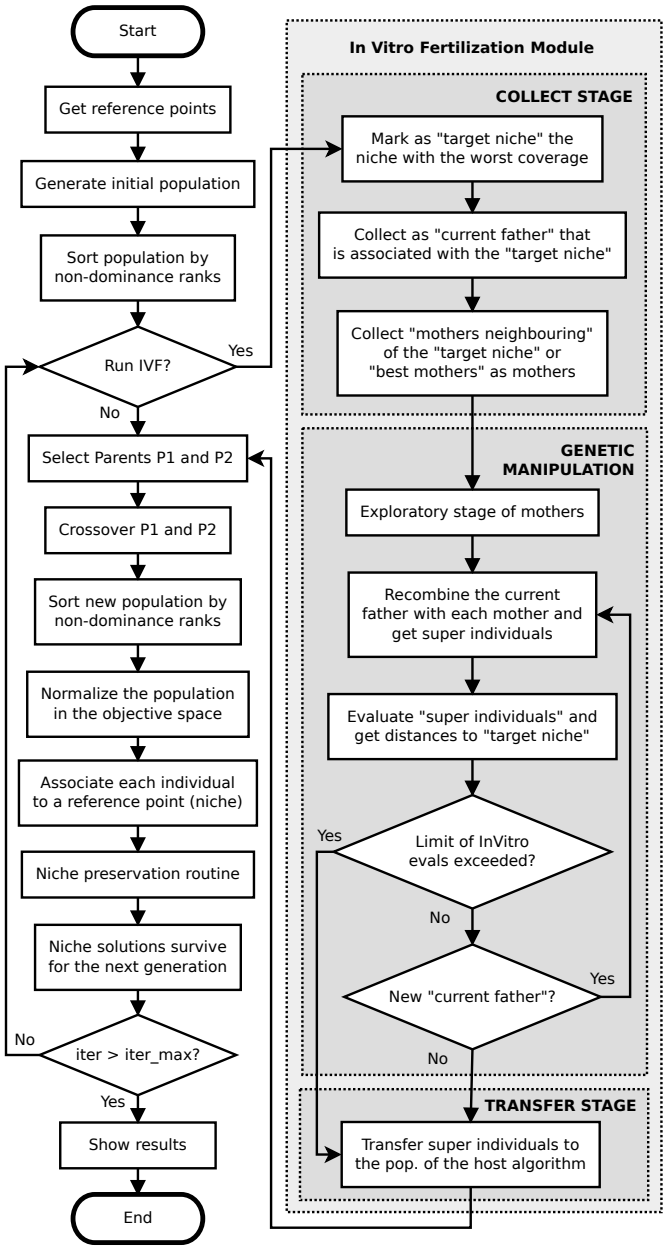

Fig. 2. IVF/NSGA-III: detailed flow, with improvements in Transfer stage.

A. Collect Stage - based on the niche with the worst coverage

As mentioned in the works [4], [5] and [6], one of the first steps of In Vitro is the Collection stage. In adapting the In Vitro Fertilization method to the many-objective approach with a strategy using reference lines and their respective niches, this proposal intends to help the host algorithm minimise the distances of solutions to their respective associated niches. In this way, the target for intervention of the auxiliary algorithm will be the niche with the worst coverage, whose individual has the greatest (worst) perpendicular distance to its respective reference line, used in the NSGA-III decomposition strategy.

This approach does the collection of the initial father based on a niche distance metric and the mothers' collection based

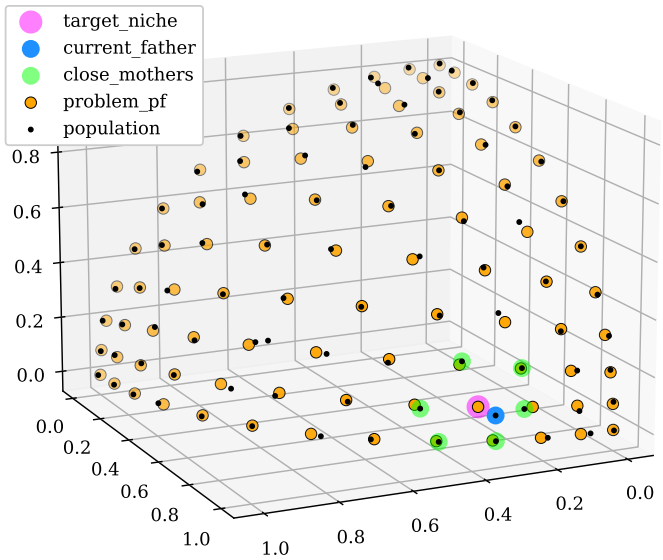

Fig. 3. Strategies for collecting mothers neighbouring the "Target Niche" which, when recombined with the "Current Father", may provide offspring closer to the "Target Niche" and intensify its result. In the caption: Problem Pareto Front (problem_pf) and Host Algorithm Population (population).

on the target niche's neighbourhood or another exploratory strategy. Fig. 3 presents an example of a population of the NSGA-III algorithm, after some generations, in treating the DTLZ-2 problem. In Fig. 3 the niche identified as the "target niche" is the one with the most distant associated solution (worst coverage). This solution is then collected as "initial current father".

The six solutions identified as "close mothers" are the other solutions closest to this niche. The number of mothers collected is a configurable parameter. This collection strategy from nearby or close mothers contributes to exploitation, and these solutions can act as beacons or lighthouses to guide the positioning of new and promising offspring solutions.

Another strategy for collecting initial mothers, which can contribute to exploration, is the selection of solutions that have a smaller distance to their respective niches, which usually occurs at the ends or borders of the approximation set. This strategy can contribute to the exploration of the search space, which can be used more in the initial stages of the search process.

B. Genetic Manipulation Stage - Differential Evolution with Beacons and Adaptive Scale Factor for Elitist Local Search

According to [12], although DE is a simple and efficient optimizer, especially for continuous optimization problems, "its structure has some limitations in the search logic, as it contains a very restricted set of exploration moves". Still, according to [12], "this fact inspired many computer scientists to improve the DE, proposing modifications in the original algorithm." Several proposals for improvements have been suggested to enhance the existing DE structure. Proposed improvements for Differential Evolution include Trigonometric

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Mutation, Adaptive Crossover Local Search, Population Size Reduction, Scale Factor Local Search, and others [12].

In line with some conclusions for improvements in the DE structure, many of them presented in [12], [13], and [14], this hybridization (IVF/NSGA-III) utilizes an elitist scheme of DE, which incorporates additional features to the standard scheme. These features include a reduced population in a promising neighborhood for local search using static beacons, a fixed Crossover Rate (CR) of 100%, an adaptive Scale Factor (F) that adjusts to the distances between a chosen beacon and the best local solution within a target niche, and the replacement of an extra perturbation vector by a normal or uniform distribution noise, also proportional to the distances between solutions and the target niche.

This proposal employs elitist DE mutation strategies, like "DE/current-to-best/0", according to (1) and Fig. 4, which is equivalent to "DE/current-to-best/1" [8], focusing on a target solution (best), without an extra random perturbation vector.

"DE/current-to-best/0":

$$\vec{V}_{i,G} = \vec{X}_{i,G} + F \cdot (\vec{X}_{best,G} - \vec{X}_{i,G}) \tag{1}$$

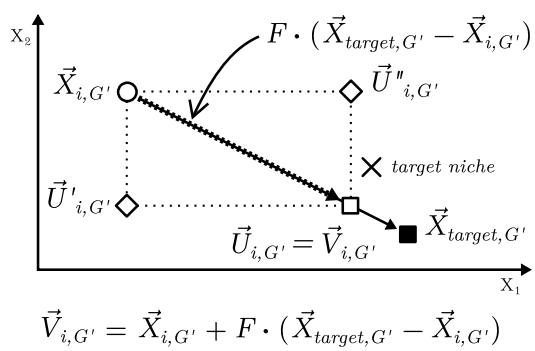

Fig. 4. Example, in 2-D search space, of the "DE/current-to-best/0" (or "DE/current-to-target/0") DE mutation strategy that will be used in this work with an adaptation of the IVF method to the Many-objective approach.

The auxiliary algorithm will use the collected subpopulation to minimize the distance to the target niche. The initial target solution (initial current father) will be considered the best solution for this local problem. Moreover, the neighbouring mothers are used as beacons to intensify this target niche. This approach can also use the Steady State method.

In this proposal, the mutation strategy "DE/current-to-best/0" will be used with a crossover rate (CR) equal to 1.0, which indicates that the recombination result will always be equivalent to the solution $\vec{V}_{i,G}$ ($\vec{U}_{i,G} = \vec{V}_{i,G}$, in Fig. 4).

Fig. 5 and 6 show how the recombination of 3 mothers and a target current father solution can obtain a descendant solution closer to the target niche. This new target solution can replace the current father in a new recombination step with the same three mothers neighbouring the target niche, and so on, according to the flow of the IVF method.

Fig. 7 (a zoom of Fig. 3 example) presents in red the result of 102 recombinations between the initial current father and the collected mothers, and also, in purple, the recombination of the new current father (second generation In Vitro) and the collected mothers, using the "DE/current-to-best/0" mutation strategy, with random $F \in \mathbb{R}, F \in [0.5, 1.0]$ and $CR = 1.0$.

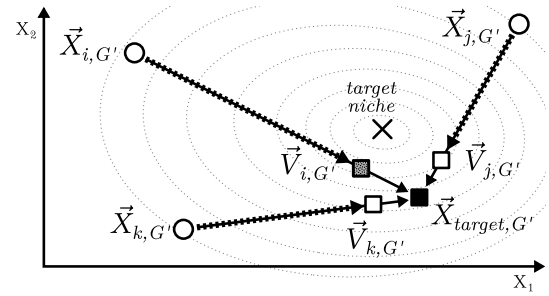

Fig. 5. Example of a first iteration of the IVF method, where three mothers ($\vec{X}_{i,G'}$, $\vec{X}_{j,G'}$ and $\vec{X}_{k,G'}$) are iterated with the initial father ($\vec{X}_{target,G'}$) and get three descendants. One of which ($\vec{V}_{i,G'}$) is closer to the target niche and will replace the current father in a subsequent iteration of IVF method.

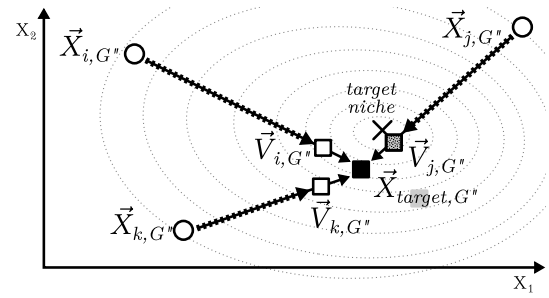

Fig. 6. Example of a subsequent In Vitro iteration where a child has outgrown the initial father and has become the new current father. A new cycle of recombination with the same neighbouring mothers is performed, this time with the new current father. Then new offspring are obtained. One of these new offspring ($\vec{V}_{j,G''}$) intensifies even closer proximity to the target niche.

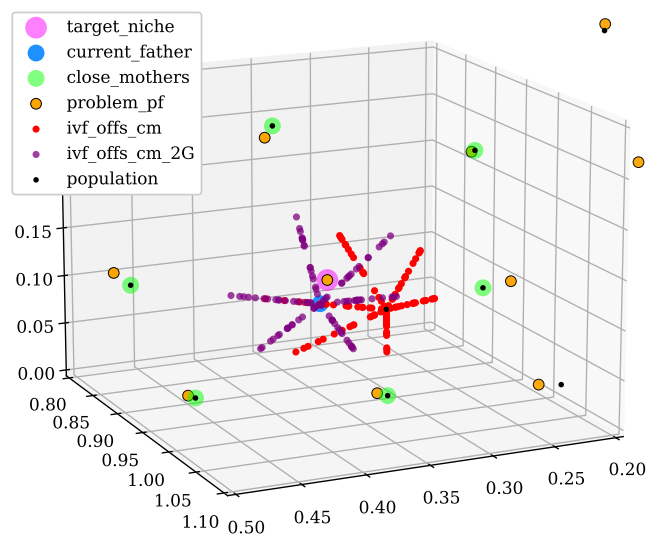

Fig. 7. In red, 102 offspring were obtained using an initial target father and six mothers neighbouring the target niche. In purple (next In Vitro generation), 102 descendants of the new current father and the same six close mothers. "DE/current-to-best/0", Random $F \in \mathbb{R}, F \in [0.5, 1.0]$ and $CR = 1.0$.

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In [15], the authors point out that, instead of choosing a fixed value between 0.0 and 2.0 for the DE mutation constant F, "recently, it was discovered that selecting F from the interval 0.5 to 1.0 randomly for each generation or each difference vector, a technique called dither, significantly improves the convergence behavior, especially for noisy objective functions". A very large F value can increase the search radius, but it can make convergence difficult [15].

Acording to [13], "several works developed various time-varying scale factor mechanisms.", and in [16] "a decrease in the scale factor value during the optimization process can lead to high quality final results."

In this work, we gradually change the scale factor (F) in the "DE/current-to-best/0" scheme to improve the approximation of new solutions over the target niche. Functions such as linear, exponential, or others that obtain intermediate positions between the current father and the iterated mother (acting as a beacon) can be used, with factors proportional to the distances between these two solutions and the target niche.

As the current father approaches the target niche, an F factor close to 1.0 can provide good offspring for the approximation to the target niche. Thus, this proposal can use a scale factor F with a value proportional to the distance between the mother and target niche and between the current father and target niche. As Fig. 8 indicates, considering "X" as the Target Niche, F can be obtained as: $F = distToX(mother)/(distToX(mother) + distToX(father))$.

Instead of decreasing the scale factor, this approach will bring it closer to 1.0 due to the structure of the DE scheme used in this implementation. The decrease will be over the difference between F and 1.0. F values greater than 1.0 can produce offspring in positions other than those between the iterated mother and the current father, resulting in a potentially exploratory advantage for the local auxiliary search. As the value of F approaches 1.0, the search radius for obtaining solutions around the current father (best) decreases. Another option is to use "DE/best-to-current/0" and approximate F towards 0.0, allowing negative values for the opposite sense.

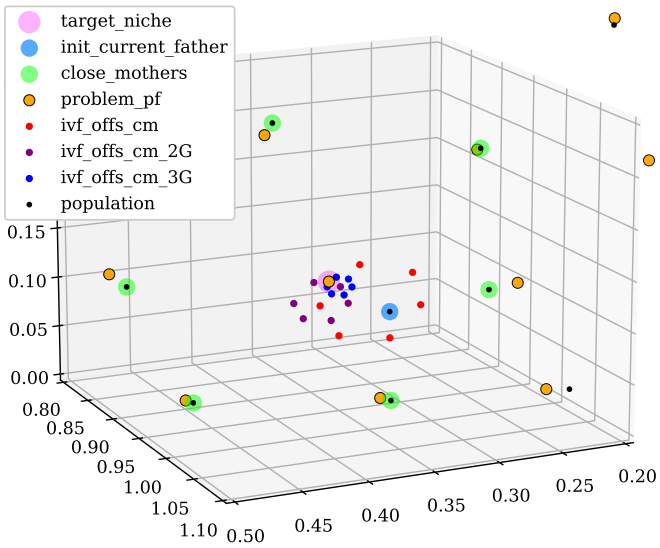

Fig. 8. Example of 3 IVF generations (red, purple and blue), each with 6 offspring obtained from the current father (of each IVF generation) and the same six close mothers ("cm"), using $CR = 1.0$ and, with "X" for Target Niche, $F = distToX(mother)/(distToX(mother) + distToX(father))$.

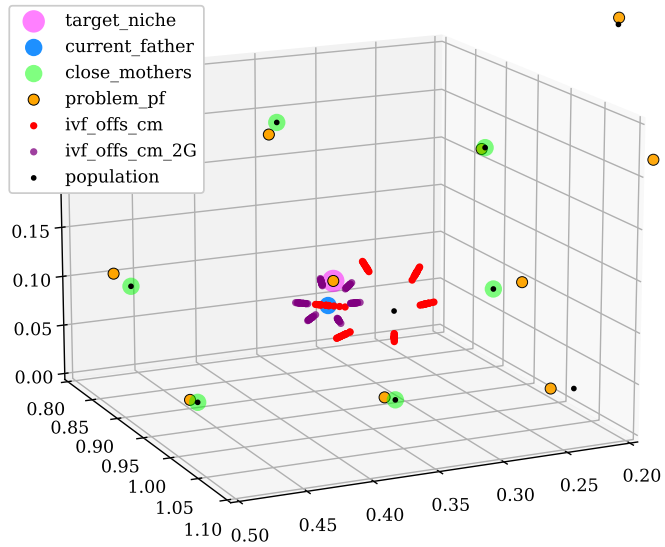

Fig. 9. Example of two IVF generations (red and purple), each with 102 offspring ("offs") obtained from the current father (of each IVF generation) and the same six close mothers ("cm"), using the same F and CR factors from Fig. 8, but including a dither with $F' = F + N(0, \sigma)$, with $\sigma = (|1 - F|)/10$.

Other strategies can be used to variate F, including a dither with a normal distribution with a σ value proportional to the distance between the current father and the target niche. Example: $F' = F + N(0, \sigma)$, with $\sigma = (|1 - F|)/10$ (Fig. 9).

C. Transfer Stage - Maintaining the niche preservation strategy of the host algorithm

Similar to the one used in [6], the transfer step will gather all the super individuals obtained in the In Vitro internal cycles in a given generation to the original population of the host algorithm. Next, the ordering and survival routine with the preservation of niches will be performed. Surplus individuals will be eliminated, and the host algorithm's population will always be the same size. The number of evaluations performed in In Vitro will be deducted from the number of descendants the host algorithm can obtain in the same generation in which In Vitro was used. In this way, each generation of the host algorithm will always have many evaluations equivalent to the size of its population, whether or not the In Vitro method has been executed.

D. Criteria for performing the In Vitro method in a generation of the host algorithm

The In Vitro Fertilization method enables different execution strategies, such as using an execution probability at each generation of the host algorithm, among other ways to reduce the consumption of evaluations. In [5] and [6], different

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execution probabilities were used, which, together with the collection size and a maximum limit of IVF internal cycles, used different amounts of evaluations, which could impact the final result. In this proposal, instead of an execution probability, a percentage limit of the current evaluations consumed by the auxiliary algorithm is defined. Thus, if IVF intensifies a target niche greatly and exceeds the configured percentage limit, it will remain idle until the host algorithm uses enough evaluations to fall below the limit again.

The gradual adaptation of the scale factor, as well as the size of the disturbance through the noise with normal distribution, both proportional to the distances of the best solution for the target niche, guided by beacons, the fixed parameter $CR = 1.0$ and the new IVF execution trigger strategy used in this work, are in line with recent research that seeks to make the algorithm parameters automatically adapt to different problems during optimization, "liberating the user from the tedious and time-consuming task of manual setting" [14].

IV. Experiments

The proposal presented in this work was implemented using the Pymoo framework, proposed by [15]. It provides several tools for evaluating the results such as the use of Hyper Volume and IGD metrics and a set of test functions also aimed at the many-objective approach, such as DTLZ and WFG problems. This experiment intends to evaluate the contribution of the collection of neighbouring mothers in their original configurations. For this reason, the AR operator of the IVF method was used, which does not alter the genes of the collected mothers. For comparison with the results of the NSGA-III algorithm itself, the same parameters described in [17] were used. DTLZ problems 1, 2, 3 and 4 were used, each with 3 and 5 objectives. The maximum number of generations used for DTLZ problems 1, 2, 3 and 4 were, respectively, 400, 250, 1000 and 600 for three objectives and 600, 350, 1000 and 1000 for five objectives. Both the NSGA-III canonical algorithm and the proposed hybrid algorithm IVF/NSGA-III were configured with the same internal NSGA-III parameters, same stop criterion and population size. The parameters CR and F, used by "DE/current-to-best/0" strategy, in IVF, in this experiment, are: $CR = 1.0$, $F' = distToX(mother)/(distToX(mother) + distToX(father))$ and $F = F' + N(0, \sigma)$, with $\sigma = (|1 - F'|)/10$, where $distToX()$ is a function to obtain perpendicular distance of a solution and the reference line of target niche.

We used the parameters of Table I from [17]: 91 reference points and 92 individuals in the population for problems with three objectives and 210 reference points and 212 individuals for problems with five objectives. We used the method Das and Dennis [18] to establish reference points, also used in [17].

As Table I indicates, eight values for "ivf_rate" were tested. For DTLZ problems 1 to 4 with three objectives, eight values for "collect_size" were tested, and, with five objectives, ten values for "collect_size". Each problem was tested 30 times to obtain IGD means and p-value calculation for statistical comparison with Wilcoxon Rank Sum Test.

TABLE I
IVF parameters used in this IVF/NSGA-III experiments

Parameter and Description	Values
IVF Rate (r): Percentage limit of evaluations used by IVF (in %)	0, 5, 7.5, 10, 12.5, 15, 20, 25, 30
Collect Size (c)**, for 3 objectives**	5, 7, 9, 12, 14, 18, 23, 28
Collect Size (c)**, for 5 objectives**	5, 7, 11, 16, 21, 27, 32, 42, 53, 64

V. Results

We present a summary of the results achieved, based on testing 576 parameter combinations (Table I). Heatmaps in Figs. 10, 11, 12, 13, 14, 15, 16 and 17 illustrate the mean IGD from 30 runs on DTLZ 1 to 4 problems with 3 and 5 objectives. Boxplots in Figs. 18 and 19 show results with ivf_rate=10% and varying ivf_collect_size for DTLZ 1 to 4 with 3 and 5 objectives. Table II shows the results breakdown with ivf_rate=10% and collect_size≈10%.

TABLE II
IGD: NSGA-III ("N") and IVF/NSGA-III ("IVF") with ivf_rate=10% and collect_size≈10% (9 if 3 obj. and 21 if 5 obj.)

DTLZ	Obj.	"N" Mean	"IVF" Mean	p-value	< 0.05
1	3	0.002309	0.001027	0.002415	✓
1	5	0.000551	0.000521	0.861213	
2	3	0.001312	0.001019	0.000097	✓
2	5	0.003529	0.003459	0.198610	
3	3	0.002392	0.000841	0.000148	✓
3	5	0.001909	0.001169	0.000529	✓
4	3	0.095390	0.095228	0.007731	✓
4	5	0.000679	0.000582	0.000241	✓

The results indicate that in problems DTLZ 1 and 2 with five objectives, there is no statistical difference between the two algorithms. In the other problems, especially complex ones, the IVF/NSGA-III intensified significantly the IGD's mean result, with a statistical difference, especially in problems DTLZ 2 with three objectives, DTLZ 3 with three and five objectives and DTLZ 4 with five objectives, with p-value < 0.001.

VI. Conclusion and future work

This work proposes a hybrid algorithm (IVF/NSGA-III) adapting the IVF module to a modified DE strategy and combining it with the NSGA-III many-objective evolutionary algorithm. The widely used DTLZ benchmark was used to validate the contribution of this approach, with DTLZ problems 1 to 4, with three and five objectives. The results indicate that this approach contributes to the intensification of solutions associated with niches, which contributes to IGD improvement. With the use of 10% of the total evaluations and with a collection of approximately 10% of the population size, the IVF auxiliary algorithm improved the results of this hybridization, outperforming the canonical NSGA-III (p-value < 0.05). In future work, we will try different many-objective benchmarks, such as DTLZ 1 to 7 and WFG, with larger objectives and variables. Furthermore, the coupling of this IVF strategy to the MOEAD algorithm (IVF/MOEAD), using Steady State, and the self-adaptation of more IVF parameters.

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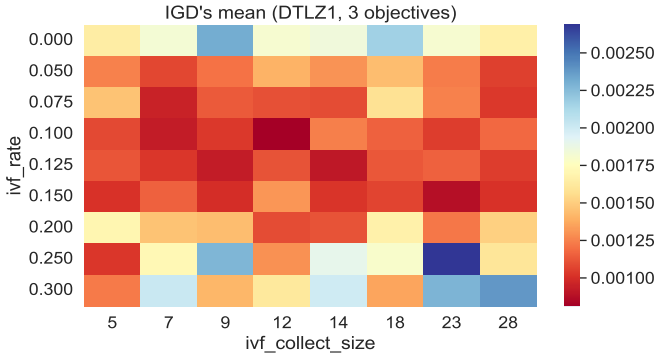

Fig. 10. Heatmap with IGD's mean, DTLZ1 problem, 3 objectives.

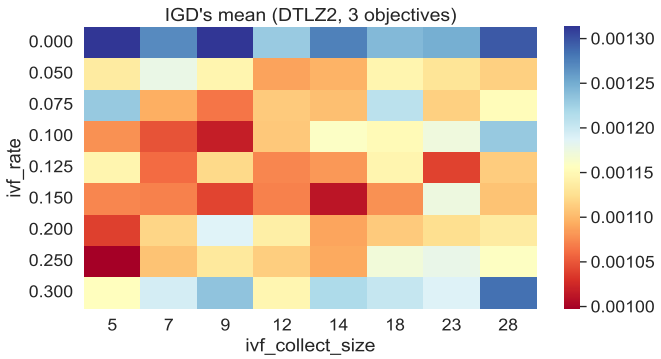

Fig. 11. Heatmap with IGD's mean, DTLZ2 problem, 3 objectives.

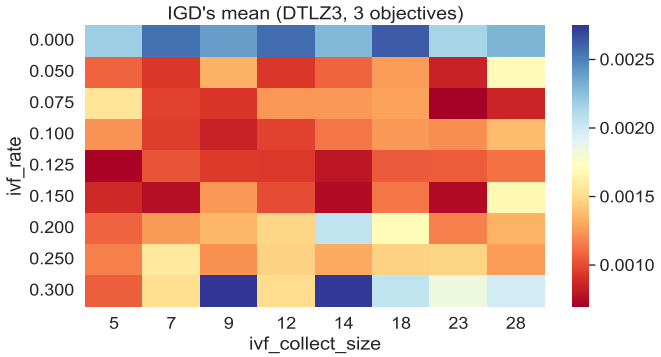

Fig. 12. Heatmap with IGD's mean, DTLZ3 problem, 3 objectives.

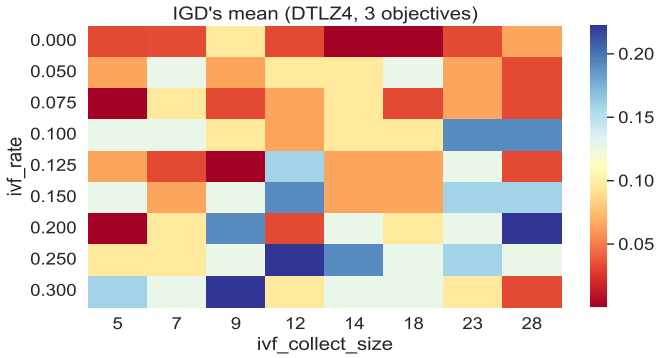

Fig. 13. Heatmap with IGD's mean, DTLZ4 problem, 3 objectives.

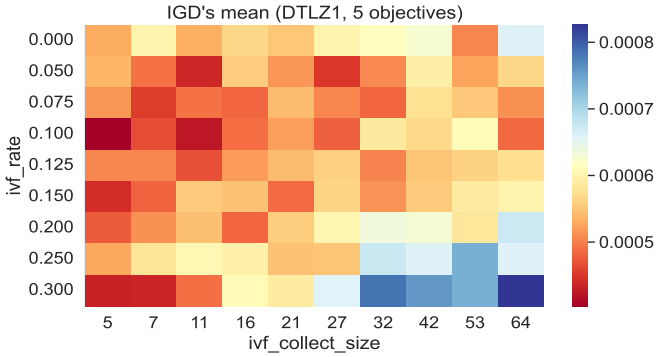

Fig. 14. Heatmap with IGD's mean, DTLZ1 problem, 5 objectives.

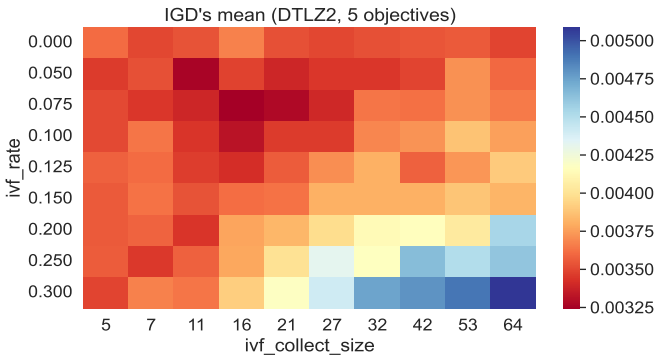

Fig. 15. Heatmap with IGD's mean, DTLZ2 problem, 5 objectives.

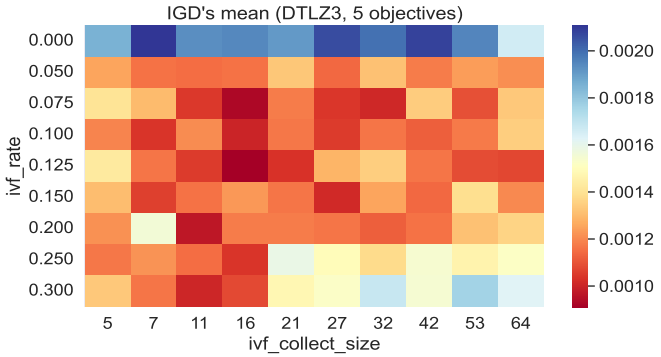

Fig. 16. Heatmap with IGD's mean, DTLZ3 problem, 5 objectives.

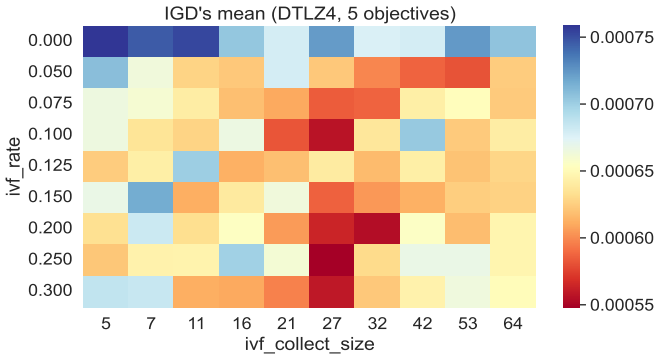

Fig. 17. Heatmap with IGD's mean, DTLZ4 problem, 5 objectives.

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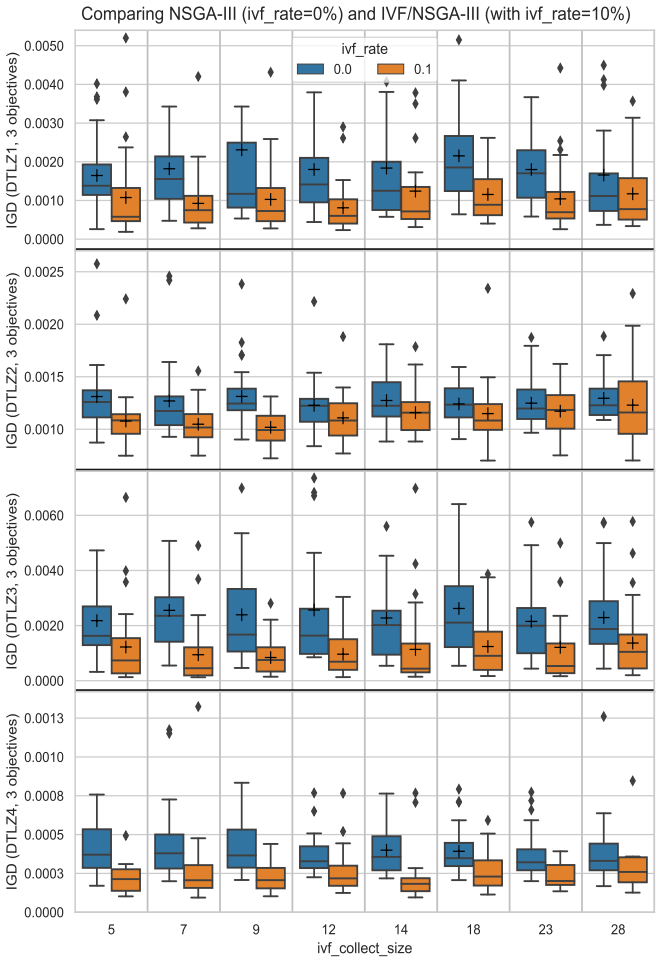

Fig. 18. IGD boxplots from DTLZ problems 1 to 4, with 3 objectives

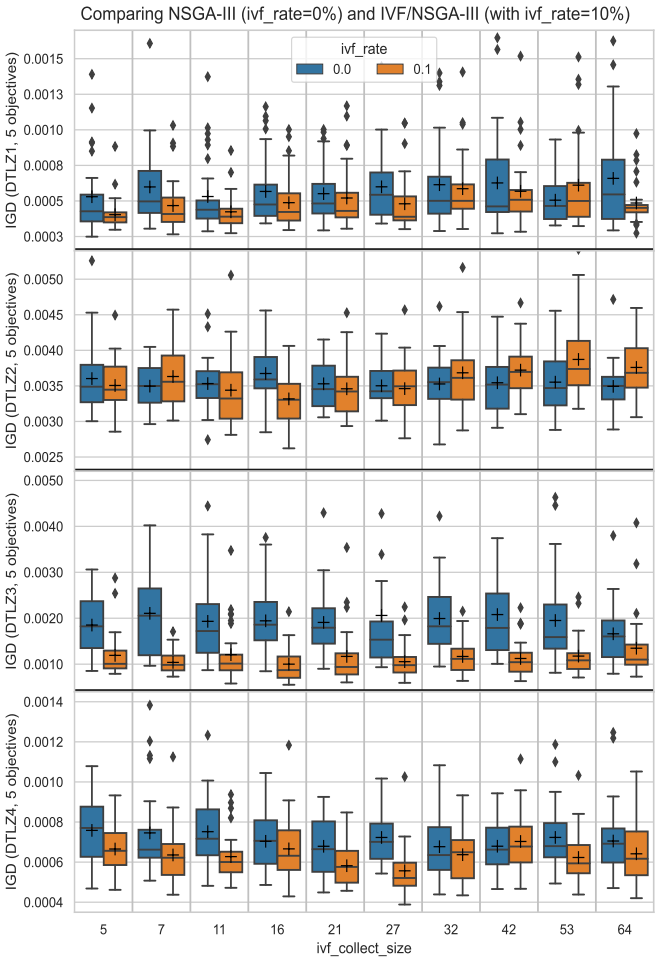

Fig. 19. IGD boxplots from DTLZ problems 1 to 4, with 5 objectives

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